INCH-POUND

MIL-E-82803(OS)
29 December 1989
SUPERSEDING
WS 22371A
21 July 1986

MILITARY SPECIFICATION

EXPLOSIVE, THIN LAYER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of thin layer explosives, referred to herein as explosive.

1.2 Classification. The explosive shall be of the following types and classes (see 6.2):

a. Type I Thin layer explosive
b. Type II Ignition thin layer explosive, Type I with the addition of MIL-I-82812 ignition strands

Class 1: 0.120-inch outside diameter
Class 2: 0.160-inch outside diameter

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-A-192	Ammonium Perchlorate, Technical
MIL-P-217	Potassium Perchlorate
MIL-A-512	Aluminum Powder, Flaked, Grained, and Atomized
MIL-P-45454	Primer, Percussion, M79 Loading, Assembling, and Packing
MIL-B-81705	Barrier Materials, Flexible, Electrostatic Protective, Heat Sealable
MIL-I-82812	Ignition Strands

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

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2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 4101 Propylene Plastic Injection and Extrusion Materials (DoD adopted)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the ignition strands shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material requirements.**

3.2.1 **Tubing.** The tubing used in the manufacture of the explosive shall be a flexible polypropylene thermoplastic tube made from material classified as PP315 E01Z and conforming to ASTM D 4101.

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3.2.1.1 **Tubing sizes.** The explosive sizes shall be as follows:

a. Type I

Outside diameter (OD): 0.120 ± 0.005 inch
Inside diameter (ID): 0.055 ± 0.005 inch

b. Type II, Class 1

OD: 0.120 ± 0.005 inch
ID: as required to accommodate three MIL-I-82812 Type I ignition strands

c. Type II, Class 2

OD: 0.160 ± 0.005 inch
ID: as required to accommodate three MIL-I-82812 Type II ignition strands

3.2.2 **Reactive material.** The reactive material on the interior wall of the explosive shall be formulated as shown in table I.

TABLE I. Reactive material formulation.

Material	Specification	Percentage by weight
Ammonium Perchlorate 1/	MIL-A-192 (Grade A)	34.0 ± 5.0
Potassium Perchlorate 2/	MIL-P-217 (Grade A)	34.0 ± 5.0
Aluminum Flake	MIL-A-512 (Grade A) Type I, Class 1	32.0 ± 5.0

1/ Weight percent includes 1.00 +1.00 -0.70 percent hydrophobic silica powder (see 6.3.1)

2/ Weight percentage includes 0.25 +0.75 -0.15 percent hydrophobic silica powder (see 6.3.1)

3.2.3 **Type II explosive ignition strand.** The ignition strand (3 each) used in the manufacture of Type II explosive shall be a material which has met the requirements of MIL-I-82812.

3.3 **Physical properties.** Physical properties of the explosive shall be as specified in table II.

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TABLE II. Physical properties.

Property	Limits	
	Minimum	Maximum
Transfer velocity, meters/second		
Type I explosive	1200	--
Type II explosive	1000	--
Tensile strength, pounds		
Type I explosive	20	--
Type II explosive	20	--
Autoignition, °C (°F), for not less than 2 hours	177 (350)	--

3.4 **Temperature and humidity (T & H).** The explosive shall function along its entire length after exposure to temperature cycling from -20 to 130°F at a relative humidity of 95 percent, minimum, for 28 days. The explosive shall also meet the transfer velocity and tensile strength requirements of table II after T & H cycling.

3.5 **Safety.**

3.5.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.6 **Shelf life.** The explosive shall have a shelf life of 10 years after date of manufacture when stored in the original, desiccated, sealed containers at -20 to 130°F.

3.6.1 **Shelf life extension.** The shelf life extension date of an individual lot may be extended for additional 1-year periods provided the explosive, upon retest, conforms to the requirements specified in table II.

3.7 **Certification.** The contractor shall certify that the explosive conforms to the requirements of 3.2.

3.8 **Workmanship.** Workmanship shall be such that the explosive is uniform in appearance and free from visible contamination.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspections (see 4.4) and the tests specified in table III performed on explosive which is representative of the production material. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

TABLE III. First article inspection.

Examinations and Tests	Requirement	Method
Autoignition	3.3	4.5.4
Temperature and humidity	3.4	4.5.5

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4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV.

TABLE IV. Quality conformance inspection.

Examination and Tests	Requirement	Method
Individual		
Visual examination	3.8 and Section 5	4.5.1
Sampling		
Transfer velocity	3.3	4.5.2
Tensile strength	3.3	4.5.3

4.4.1 **Lot.** A lot shall consist of all the explosive manufactured in one continuous run, using the same lots of materials, and made by the same processes and procedures, and offered to the procuring activity for acceptance at one time.

4.4.2 **Sampling.** Sampling shall consist of a minimum of five feet of explosive cut from both ends of the material.

4.5 **Test methods.** The following test methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to 3.8 and Section 5.

4.5.2 **Transfer velocity.** A 3-foot (minimum) segment of explosive shall be initiated using a MIL-P-45454 M79 primer, or equivalent, and the transfer velocity shall be measured.

4.5.3 **Tensile strength.** A 2-foot (minimum) segment of explosive shall be subjected to a 25 $\pm$ 3 pound static load for 1 minute minimum. If the explosive fails during the test, perform an additional tensile strength test on a new length of explosive using a tensile testing machine to determine actual tensile strength.

NOTE: Second test is not required if a tensile testing machine was used during first test and actual tensile strength value was obtained.

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4.5.4 **Autoignition.** A 6-inch (minimum) length of explosive shall not ignite when exposed to a temperature of 177 to 182°C (350 to 360°F) for 2 hours minimum. The explosive shall be placed in a container preconditioned to the required temperature and maintained at the temperature for the required time.

4.5.5 **Temperature and humidity (T & H).** Twenty segments of explosive, each a minimum of 2 feet long, with the tube ends sealed closed shall be tested as follows:

a. Subject 17 of the explosive samples to the test of MIL-STD-331, Method 105.1, Two Chamber Method, except that the temperature range shall be -20 to 130°F.

b. Subject the three remaining explosive samples (controls) to a room temperature environment for the duration of the test specified above.

c. Within 24 hours of the conclusion of the test period, separate and condition the samples as specified below:

(1) Six T & H specimens and one control specimen shall be conditioned to 40 ± 10°F for a minimum of 1 hour.

(2) Five T & H specimens and one control specimen shall be conditioned to 75 ± 10°F for a minimum of 1 hour.

(3) Six T & H specimens and one control specimen shall be conditioned to 100 ± 10°F for a minimum of 1 hour.

d. At the end of the conditioning period remove the end seal, or seals, from the specimens and initiate the test specimens, while at the conditioning temperature, in accordance with the following:

(1) Type II explosive specimens shall be initiated using a MIL-P-45454 M79 primer, or equivalent. Visually examine the samples for charring to ensure that functioning occurred along the entire length.

(2) Tape a series of Type I explosive specimens together that have been conditioned at the same temperature, then tape to a 2.0 ± 0.4 inch length of Type II explosive that has been closed at the untaped end by heat sealing. Initiate the Type I explosive using a MIL-P-45454 M79 primer, or equivalent. Also measure the transfer velocity of two or more lengths of the Type I explosive that has been subjected to the T & H test (see 4.5.5). Subject a length of the Type I explosive that has been subjected to the T & H test to the tensile strength test (see 4.5.3). Visually verify that the Type I explosive ignited the attached Type I or Type II explosive.

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4.6 **Inspection of packaging.** The explosive containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C.

5.1.1 **Preservation.** Preservation of the explosive shall be in accordance with MIL-P-116, submethod IA-8, with desiccant. The bag material shall be in accordance with MIL-B-81705.

5.1.2 **Packing.** The explosive shall be packed to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packing shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation. Care shall be taken when packing type II explosive to ensure that the ignition strands are not free to move.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Shelf life and inspection/test date for extension (see 3.6). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 1-YEAR PERIOD PROVIDED THE EXPLOSIVE, UPON RETEST, CONFORMS TO THE REQUIREMENTS SPECIFIED IN TABLE II OF MIL-E-82803."

e. Storage conditions.

f. Toxicity warning (if applicable).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The explosive specified herein is intended for use in the manufacture of the Standard Missile Mk 104 Dual Thrust Rocket Motor igniters.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Type and class required (see 1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

d. Whether first article inspection is required (see 3.1 and 6.4).

e. Material Safety Data Sheets required (see 3.5.2 and 6.5).

f. DOD FAR 52.246-15 (see 3.7).

g. Packaging required, if different than 5.1.

h. Special marking, if other than as specified in 5.2.2.

i. Safety precautions required (see 6.6).

6.3 Possible materials. The following materials have been found to meet the requirements of this specification. This list is given for information only.

6.3.1 Silica powder. Aerosil R972 manufactured by Degussa Corporation, Peterboro, New Jersey, has been used in the explosive reactive material.

6.3.2 Typical material. Thin layer explosive and ignition thin layer explosive manufactured by Explosive Technology, Fairfield, CA, have been found to meet the requirements of this specification.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and

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disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1376-N363)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-82803	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
EXPLOSIVE, THIN LAYER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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